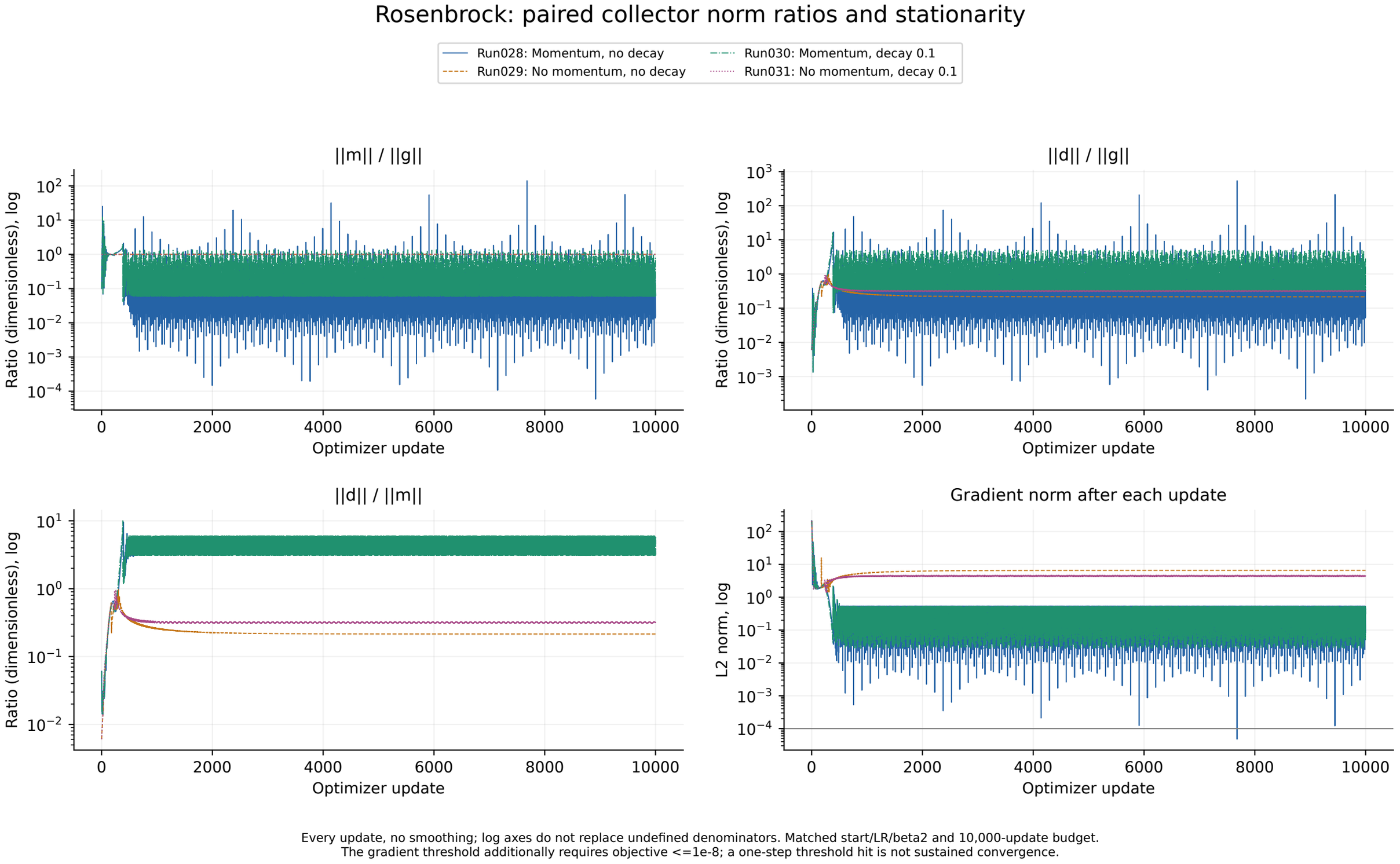

Rosenbrock: paired collector norm ratios and stationarity
Run028: Momentum, no decay
Run029: No momentum, no decay
Run030: Momentum, decay 0.1
Run031: No momentum, decay 0.1
	m		/		g	
	d		/		g	
	d		/		m	
Gradient norm after each update
Ratio (dimensionless), log
L2 norm, log
Optimizer update
Every update, no smoothing; log axes do not replace undefined denominators. Matched start/LR/beta2 and 10,000-update budget.
The gradient threshold additionally requires objective <=1e-8; a one-step threshold hit is not sustained convergence.